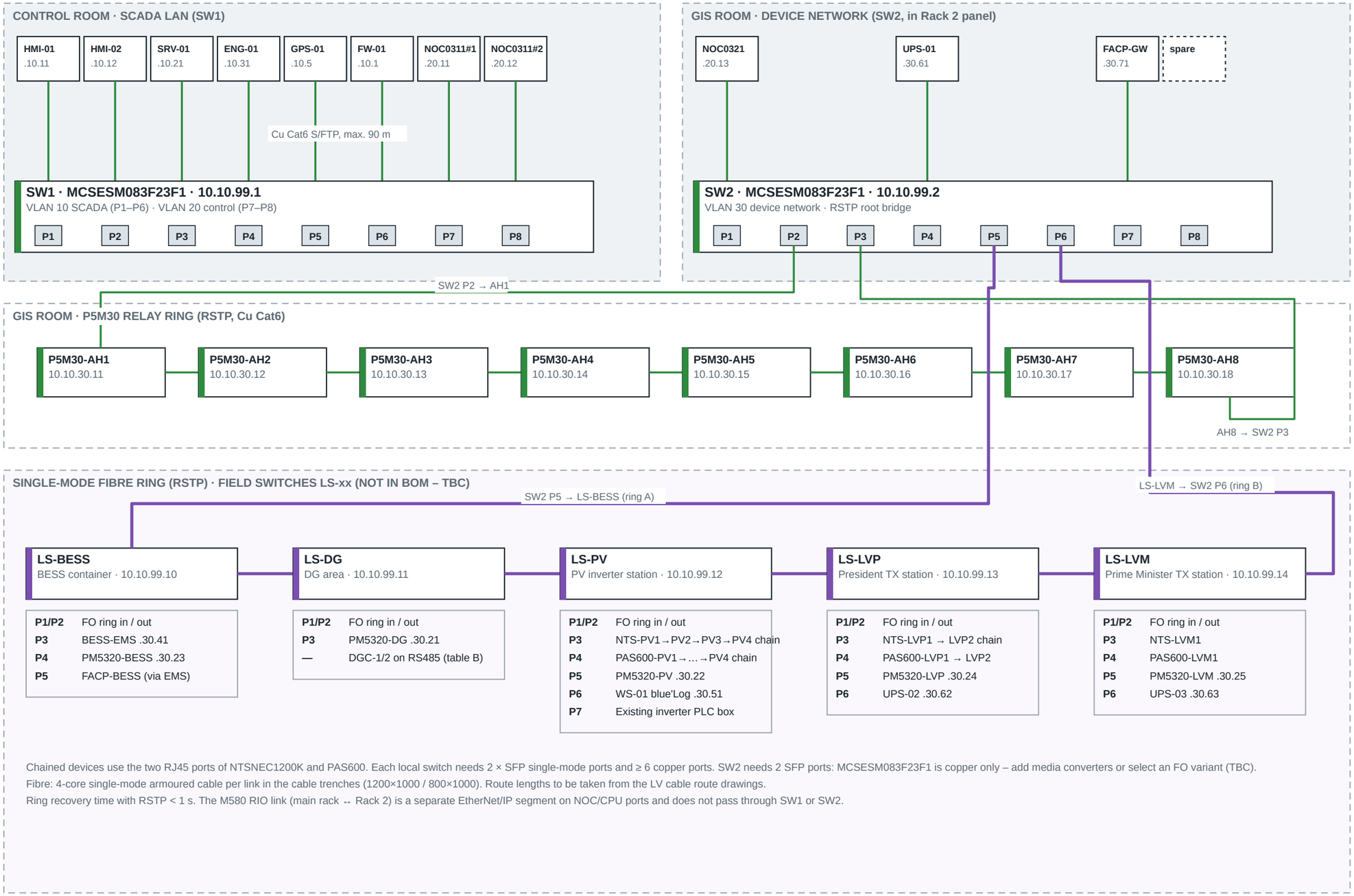


P1/P2	FO ring in / out
P3	BESS-EMS .30.41
P4	PM5320-BESS .30.23
P5	FACP-BESS (via EMS)

P1/P2	FO ring in / out
P3	PM5320-DG .30.21
—	DGC-1/2 on RS485 (table B)

P1/P2	FO ring in / out
P3	NTS-PV1→PV2→PV3→PV4 chain
P4	PAS600-PV1→…→PV4 chain
P5	PM5320-PV .30.22
P6	WS-01 blue'Log .30.51
P7	Existing inverter PLC box

P1/P2	FO ring in / out
P3	NTS-LVP1 → LVP2 chain
P4	PAS600-LVP1 → LVP2
P5	PM5320-LVP .30.24
P6	UPS-02 .30.62

P1/P2	FO ring in / out
P3	NTS-LVM1
P4	PAS600-LVM1
P5	PM5320-LVM .30.25
P6	UPS-03 .30.63

Chained devices use the two RJ45 ports of NTSNEC1200K and PAS600. Each local switch needs 2 × SFP single-mode ports and ≥ 6 copper ports. SW2 needs 2 SFP ports: MCSESM083F23F1 is copper only – add media converters or select an FO variant (TBC).
Fibre: 4-core single-mode armoured cable per link in the cable trenches (1200×1000 / 800×1000). Route lengths to be taken from the LV cable route drawings.
Ring recovery time with RSTP < 1 s. The M580 RIO link (main rack ↔ Rack 2) is a separate EtherNet/IP segment on NOC/CPU ports and does not pass through SW1 or SW2.

TABLE A – VLAN AND IP ADDRESS PLAN

VLAN	Subnet	Use	Gateway / routing
10	10.10.10.0/24	SCADA LAN (HMI, server, ENG)	FW-01 .1 (to DMZ only)
20	10.10.20.0/24	M580 control (CPU, NOC, CRA)	None
30	10.10.30.0/24	Field devices	BMENOC0321 IP forwarding
99	10.10.99.0/24	Switch management	ENG-01 only

Device	IP address	Remark
FW-01 / GPS-01	10.10.10.1 / .5	NTP server for all devices
HMI-01 / HMI-02 / SRV-01 / ENG-01	.10.11 / .12 / .21 / .31	Static addresses
M580 CPU / NOC0311 #1 / #2 / NOC0321	10.10.20.10 / .11 / .12 / .13	NOC0321 also 10.10.30.1
BMECRA31210 (Rack 2)	10.10.20.20	RIO drop
NTS-PV1…PV4 / LVP1, LVP2 / LVM1	10.10.30.101–104 / 105, 106 / 107	EtherNet/IP adapters
P5M30-AH1 … AH8	10.10.30.11 … 18	Modbus TCP, unit ID 1
PM5320 DG / PV / BESS / LVP / LVM	10.10.30.21 … 25	Modbus TCP
PAS600-PV1…4 / LVP1, 2 / LVM1	10.10.30.31 … 37	Modbus TCP gateway for sensors
BESS-EMS / WS-01 / UPS-01…03 / FACP-GW	.30.41 / .51 / .61–.63 / .71	Modbus TCP

TABLE B – RS485 MODBUS RTU BUSES

Bus	Devices (Modbus ID)	Settings	Remark
NOM0200 port 1	DGC-1 (1), DGC-2 (2)	19200 8N1	Shielded twisted pair, 120 Ω ends
NOM0200 port 2	Spare	—	Reserved
NTSFMB0120K PV1	INV NB01 (1)	9600 8N1	Point-to-point, see note 4
NTSFMB0120K PV2	INV NB02 (1)	9600 8N1	Point-to-point
NTSFMB0120K PV3	INV NB03 (1)	9600 8N1	Point-to-point
NTSFMB0120K PV4	INV NB04 (1)	9600 8N1	Point-to-point
NTSFMB0120K LV	Spare	—	Future meters

TABLE C – CABLE TYPES

Type	Specification
Ethernet Cu	Cat6 S/FTP industrial, RJ45 shielded, ≤ 90 m
Fibre	Single-mode OS2, 4-core armoured, LC duplex, 1310 nm
RS485	2-pair shielded twisted pair 120 Ω (e.g. Belden 3106A), shield earthed at master end
Hard-wired I/O	Multicore 1.0 mm² screened, 24 V DC, separate from power cables

LEGEND

- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- RS485 – Modbus RTU
- EtherNet/IP (M580 RIO / DIO)
- Switch port Pn

NOTES

1. Read with sheet 001. Addresses are proposals for approval; final values to be fixed at FAT.
2. All switches managed: unused ports shut down, port security (MAC binding), SNMPv3, syslog to SRV-01, RSTP with SW2 as root bridge.
3. IP forwarding on BMENOC0321 is restricted to SCADA ↔ device traffic; no other routing between VLANs except via FW-01.
4. Communication schematic BFE/PR/26/01 shows the 4 inverters daisy-chained to one PLC box. The offer provides one NTSFMB0120K per PV area, so point-to-point links are proposed for faster polling and fault isolation (TBC).
5. Field switches LS-xx, SFP ports on SW2 and fibre cables are additions to the BOM (TBC).
6. All devices take time from GPS-01 via NTP; P5M30 event time-stamps at 1 ms.

REV	DESCRIPTION	DATE
A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
COMMUNICATION NETWORK
Topology, ports, IP plan and serial buses

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-002** **REV A**